Networks, Signals and Systems

Quiz 1 : Answer Key

Q1. $y(t) = \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} x(\tau)\, d\tau$.

a) A system is said to be an LTI system if it satisfies the following properties.

1/2 i) Linearity: A system is said to be linear if it follows the principle of superposition, ie, if $y_1(t)$ is the o/p of $x_1(t)$ and $y_2(t)$ is the o/p of $x_2(t)$, then the o/p of $ax_1(t) + bx_2(t)$ is $ay_1(t) + by_2(t)$, for any constant $a, b \in \mathbb{C}$

1/2 ii) Time Invariance: A system is said to be time invariant if $y(t)$ is the o/p of $x(t)$, then $y(t-t_0)$ is the o/p of $x(t-t_0)$.

To prove linearity,

o/p of $x_1(t)$: $y_1(t) = \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} x_1(\tau)\, d\tau$

o/p of $x_2(t)$: $y_2(t) = \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} x_2(\tau)\, d\tau$

o/p of $ax_1(t) + bx_2(t)$: $y_3(t) = \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} (ax_1(t) + bx_2(t))\, dt$

$$= \frac{a}{2\Delta} \int_{t-\Delta}^{t+\Delta} x_1(t)\, dt + \frac{b}{2\Delta} \int_{t-\Delta}^{t+\Delta} x_2(t)\, dt$$

$$= a y_1(t) + b y_2(t)$$

$\Rightarrow$ o/p of $ax_1(t) + bx_2(t) = ay_1(t) + by_2(t)$

$\therefore$ The system is linear $\longrightarrow$ 1 mark for steps, 1/2 for final statement.

To prove time invariance,

Let $y_1(t)$ be the o/p of $x(t)$ $\Rightarrow$ $y_1(t) = \frac{1}{2\Delta}\int_{t-\Delta}^{t+\Delta} x(\tau)\,d\tau$

Shifting $y_1(t)$ by t_0, we get,

$$y_1(t-t_0) = \frac{1}{2\Delta}\int_{t-t_0-\Delta}^{t-t_0+\Delta} x(\tau)\,d\tau$$

Let $y_2(t)$ be the o/p of $x(t-t_0)$, then,

$$y_2(t) = \frac{1}{2\Delta}\int_{t-\Delta}^{t+\Delta} x(\tau-t_0)\,d\tau$$

Let $\alpha = \tau - t_0$, then $d\alpha = d\tau$, and the limits become,

$\tau = t+\Delta \Rightarrow \alpha = t - t_0 + \Delta$

$\tau = t-\Delta \Rightarrow \alpha = t - t_0 - \Delta$

$$\Rightarrow y_2(t) = \frac{1}{2\Delta}\int_{t-t_0-\Delta}^{t-t_0+\Delta} x(\alpha)\,d\alpha$$

Using change of variable $\alpha = \tau$,

$$\Rightarrow y_2(t) = \frac{1}{2\Delta}\int_{t-t_0-\Delta}^{t-t_0+\Delta} x(\tau)\,d\tau = y_1(t-t_0)$$

Shifting i/p and shifting o/p give the same expression.

$\therefore$ The system is time invariant. $\longrightarrow$ 1 mark for steps, 1/2 for final statement.

b) Causal: if the output $y(t)$ doesn't depend on future values of input $[x(t)]$ — 1/2

Stability: BIBO stable ⇒ for every bounded input, output would be bounded — 1/2

⇒ $|x(t)| \leq M \ \forall t$ & $|y(t)| \leq M_2 \ \forall t$ [optional]

$$y(t) = \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} x(\tau)\, d\tau \quad , \ \Delta > 0$$

$y(t)$ is affected by $x(t+\alpha)$ $\alpha \in [0, \Delta]$

⇒ $y(t)$ is dependent on future as well and since $\Delta > 0$, system is not causal. — 1

for any bounded input $x(t)$ ⇒ $|x(t)| < M_1 \ \forall \ t \in \mathbb{R}$

$$\Rightarrow |y(t)| = \left| \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} x(\tau)\, d\tau \right| \Rightarrow |y(t)| \leq \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} |x(t)|\, d\tau \quad \text{— 1/2}$$

$$\leq \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} M_1 \, d\tau$$

$t-\Delta$

$$|y(t)| \leq \frac{1}{2\Delta} \int_{t-\Delta}^{t+\Delta} M \, d\tau$$

$$\Rightarrow |y(t)| \leq \frac{1}{2\Delta} \cdot M_1 \cdot 2\Delta \Rightarrow |y(t)| \leq M_1$$

$\Rightarrow$ $y(t)$ is also bounded - ✓

$\Rightarrow$ System is stable

1) c) **Method 1:** [Total : 3M]

Given,

$$y(t) = \frac{1}{2\Delta}\int_{t-\Delta}^{t+\Delta} x(\tau)\,d\tau, \quad \Delta > 0$$

→ $y(t) = u(t)$ when $x(t) = \delta(t)$ → Substitute in above equation → 1M

$$u(t) = \frac{1}{2\Delta}\int_{t-\Delta}^{t+\Delta} \delta(\tau)\,d\tau$$

→ using sifting property of the Dirac-Delta function we can say,

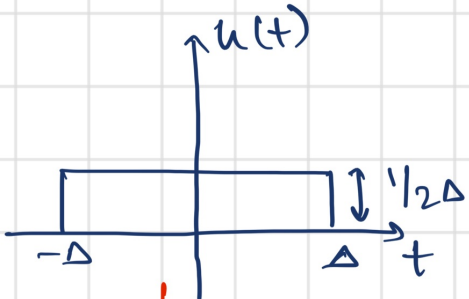
$$\int_{t-\Delta}^{t+\Delta} \delta(\tau)\,d\tau = 1 \quad \text{if} \quad t+\Delta \geq 0 \ \&\ t-\Delta \leq 0$$
$$\Rightarrow t \geq -\Delta \ \&\ t \leq \Delta$$

→ 1M

Therefore for all $t \in \mathbb{R}$, the impulse response will be,

$$u(t) = \begin{cases} \frac{1}{2\Delta} & \text{if } -\Delta \leq t \leq \Delta \\ 0, & \text{otherwise} \end{cases}$$

→ 1M



optional

Method 2:

→ $$u(t) = \frac{1}{2\Delta}\int_{t-\Delta}^{t+\Delta} \delta(\tau)\,d\tau$$ → 1M

$$= \frac{1}{2\Delta}\left[\int_{-\infty}^{t+\Delta} \delta(\tau)\,d\tau - \int_{-\infty}^{t-\Delta} \delta(\tau)\,d\tau\right] \quad \text{but} \int_{-\infty}^{t} \delta(\tau)\,d\tau = u(t)$$

↓ Unit-step impulse

$$u(t) = \frac{1}{2\Delta}\left(u(t+\Delta) - u(t-\Delta)\right) \ \forall\, t \in \mathbb{R}$$ → 0.5M

But,

→ $u(t+\Delta) = 1$ if $t+\Delta \geq 0$

$t \geq -\Delta$

→ $u(t-\Delta) = 1$ if $t-\Delta \geq 0$

$t \geq \Delta$

So for $u(t+\Delta) - u(t-\Delta) = 1$ we want $u(t+\Delta) = 1$ & $u(t-\Delta) = 0$

$\therefore$ $t+\Delta \geq 0$ & $t-\Delta \leq 0$

$t \geq -\Delta$ & $t \leq \Delta$ → combining both we get

$-\Delta \leq t \leq \Delta$ → 0.5M

$$\therefore u(t) = \begin{cases} \frac{1}{2\Delta} & \text{if } -\Delta \leq t \leq \Delta \\ 0, & \text{otherwise} \end{cases} \quad \forall\, t \in \mathbb{R}$$

→ 1M

Exam..... Subject: .....

(2)(a) Given $x(t)$ is periodic → Let T_0 be it's fundamental period.

$\therefore x(t+nT_0) = x(t)$ $\forall n \in Z$ → 0.5M

$y(t) = x(100-7t)$

$y(t+T_0) = x(100-7(t+T_0))$

$= x(100-7t-7T_0)$

$= x((100-7t)+(-7)T_0) = x(100-7t)$

$= y(t)$ → proving periodic 0.5M

$\therefore T_0$ is the period of $y(t)$ and $y(t)$ is periodic.

For fundamental period,

$y(t+T) = x(100-7t-7T)$

We need to shift $x(t)$ least by T_0 so that it just reaches next cycle. → 0.5M for correct reason

$\therefore$ Shift $= 7T = T_0 \Rightarrow T = T_0/7$. → 0.5M for correct answer

ie., we need to shift $y(t)$ least by $T_0/7$ to get same value.

$\therefore T_0/7$ is the fundamental period of $y(t)$.

(Any other method will fetch full if proved properly)

Q.2(b) $\Rightarrow$ $x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$ ——— (0.5)

$\Rightarrow$ $y(t) = x(100 - 7t)$

$= \sum a_n e^{jn\omega_0 (100 - 7t)}$

$= \sum a_n \cdot e^{jn\omega_0 (100)} \cdot e^{-jn\omega_0 (7t)}$ ——— (1)

$\Rightarrow$ And the Time period of $y(t) = \frac{T_0}{7}$

where T_0 is the Time period of $x(t)$

$\Rightarrow$ $\therefore$ ang. freq. of $y(t) = \frac{2\pi}{T_0/7} = 7\left(\frac{2\pi}{T_0}\right)$

& $\frac{2\pi}{T_0} = \omega_0$ (angular freq. of $x(t)$)

$\therefore$ $= 7\omega_0$

$\Rightarrow$ $\therefore$ $y(t) = \sum a_n e^{jn\omega_0 (100)} \cdot e^{-jn(7\omega_0)t}$ ——— (1)

Now let $m = -n$

$\Rightarrow$ So, $y(t) = \sum (a_{-m} e^{-jm\omega_0 (100)} \cdot) \cdot e^{jm(7\omega_0)t}$

↳ FS representation ——— (0.5)

$\Rightarrow$ FS coefficient of $y(t) = a_{-m} e^{-jm\omega_0 (100)}$